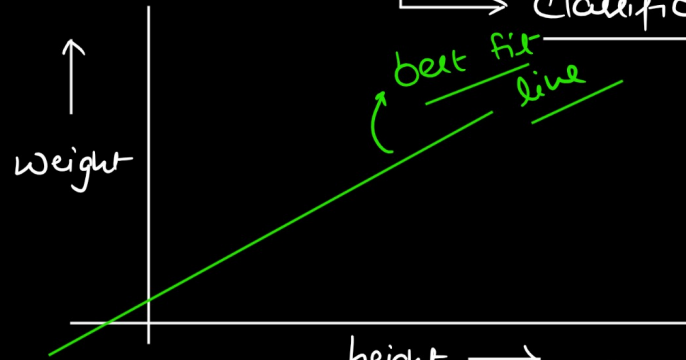


Logistic Regression (Binary Classification)

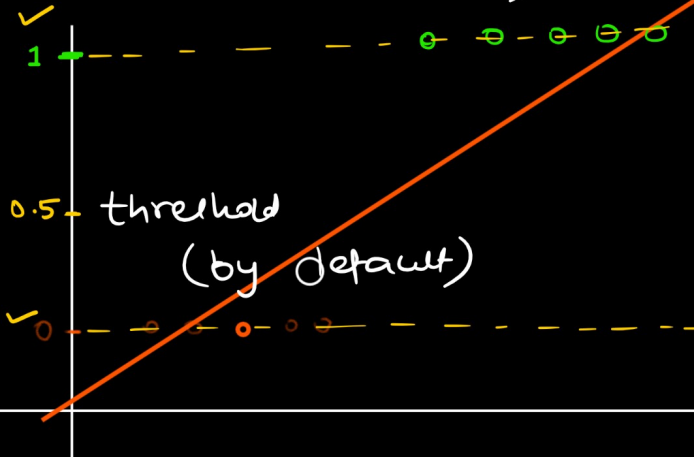


$\hat{y} \rightarrow$ Predicted value (Model Prediction)

$y \rightarrow$ Actual value (dataset)

$$\hat{y} = mx + c$$

new
test
Data



Probability

$$\rightarrow 0.794$$

Diabetic Classification

target (outcome)

Sigmoid Function

$$\text{Probability} = \frac{1}{1 + e^{-\hat{y}}}$$

$f(\hat{y})$

1 $\rightarrow$ diabetic

0

0 $\rightarrow$ not

1

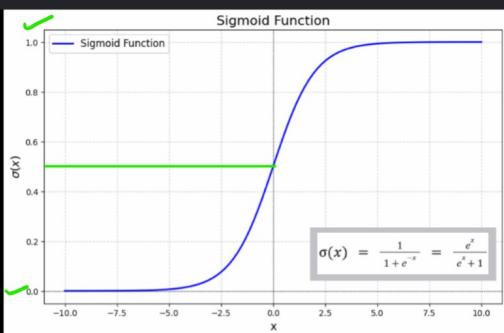
diabetic

0

0

0

1





Probability

Sigmoid function

$$\sigma(\hat{y}) = f(\hat{y}) = \frac{1}{1 + e^{-\hat{y}}}$$

$\hat{y} = mx + c$

binary

classification

task

$$f(\hat{y}) = \frac{1}{1 + e^{-(mx + c)}}$$

$$f(\hat{y}) = \frac{1}{1 + \frac{1}{e^{mx + c}}}$$

$$f(\hat{y}) = \frac{1}{\frac{e^{mx + c} + 1}{e^{mx + c}}}$$

$$P(A) = f(\hat{y}) = \frac{e^{mx + c}}{e^{mx + c} + 1}$$

Binary classification

$$P(A) + P(B) = 1$$

$$P(B) = 1 - P(A)$$

$$P(B) = \frac{e^{mx+c}}{e^{mx+c} + 1}$$

$$P(B) = \frac{\cancel{e^{mx+c}} + 1 - \cancel{e^{mx+c}}}{e^{mx+c} + 1}$$

$$P(B) = \frac{1}{e^{mx+c} + 1}$$

$$\frac{P(A)}{P(B)} = \frac{\frac{e^{mx+c}}{e^{mx+c} + 1}}{\frac{1}{e^{mx+c} + 1}} = e^{mx+c}$$

$$\frac{P(A)}{P(B)} = \frac{\sigma(\hat{y})}{1 - \sigma(\hat{y})} = e^{mx+c}$$

$$\log_e \left(\frac{P(A)}{P(B)} \right) = \log_e e^{mx+c}$$

$$\log_e \left(\frac{P(A)}{P(B)} \right) = mx+c \cdot \log_e e$$

$$\log_e \left(\frac{P(A)}{P(B)} \right) = mx+c = \hat{y}$$

$$P(A) = 0.8$$

$$P(B) = 0.2$$

$$\log_e \left(\frac{P(A)}{P(B)} \right) = \hat{y}$$

→ Prediction
by the
model

$$\log_e \left(\frac{0.8}{0.2} \right) = \hat{y}$$

→ A
+ve class

$$1.386 = \hat{y} \rightarrow \text{+ve value}$$

→ B

-ve class

→ -ve value

$$P(B) = 0.8$$

$$P(A) = 0.2$$

$$\log_e \left(\frac{0.2}{0.8} \right) = \hat{y} = -1.386$$

xx

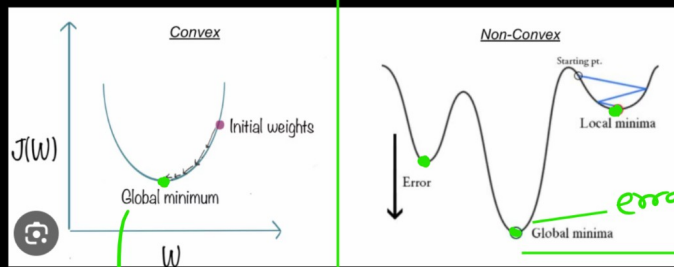
Classification (Predicted value)

→ 0, 1

$$\text{Cost function} = \text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Non-convex
curve

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n \left(y_i - \frac{1}{1 + e^{-(mx+c)}} \right)^2$$



Gradient descent optimizer

Diabetic local, global minima (easy)

$$y=1 \left\{ \begin{array}{l} \text{Error} \downarrow \downarrow \\ \hat{y}=1 \end{array} \right\} \xrightarrow{\text{Wrong Classification}} \begin{array}{l} y=1 \\ \hat{y}=0 \end{array} \xrightarrow{\text{Error}} -\log_e(0) = \frac{\text{undefined}}{\infty}$$

$$-\log_e(1) = 0$$

Non-diabetic

cost function $(y, \hat{y}) =$

$$\left\{ \begin{array}{l} -\log_e(\hat{y}) \\ -\log_e(1-\hat{y}) \end{array} \right. \begin{array}{l} y=1 \\ y=0 \\ (1-y) \end{array}$$

Standard Reference

convex curve

$$\text{Cost-Function}(y, \hat{y}) = \sum_{i=1}^n -y_i \log_e \hat{y}_i - (1-y_i) \log_e (1-\hat{y}_i)$$

$$\text{Error} \uparrow \uparrow \left\{ \begin{array}{l} y=0 \\ \hat{y}=1 \end{array} \right\} \xrightarrow{\text{Misclassification}} -\log_e(1-1) = -\log_e(0) = \infty$$

$$\text{Error} \downarrow \downarrow \left\{ \begin{array}{l} y=0 \\ \hat{y}=0 \end{array} \right\} \xrightarrow{\text{Correct Classification}} -\log_e(1-0) = -\log_e 1 = 0$$

Evaluation (wrt Prediction)

Confusion Matrix

		Actual	
		(1)	(0)
Predicted	Positive (1)	TP	FP
Negative (0)	FN	TN	

100 Records → 95 Records → non-diabetic (95%)
5 Records → diabetic (5%)

① Data Imbalanced

↳ don't prefer to go for accuracy

$$\text{Accuracy} \rightarrow \frac{TP + TN}{TP + TN + FP + FN}$$

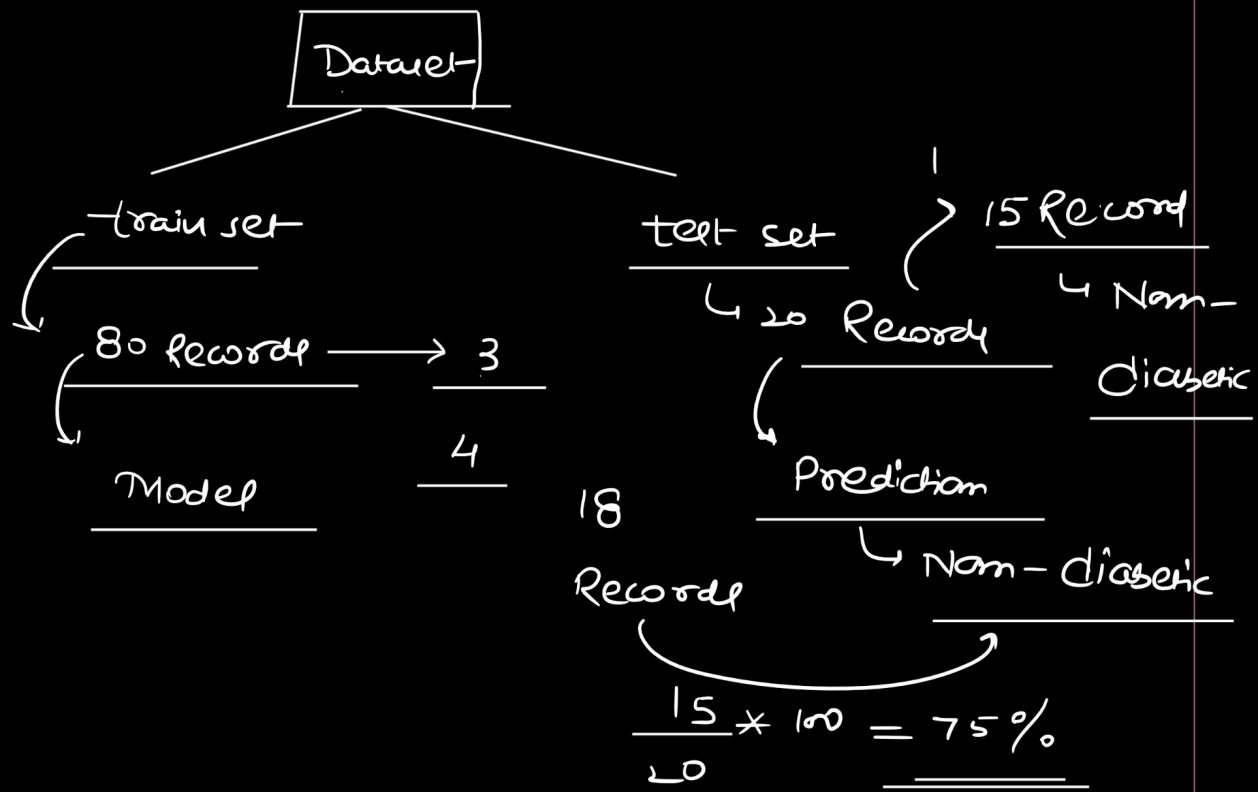
$$\text{Recall} \rightarrow \frac{TP}{TP + FN}$$

↑↑ health case
actual → 1
prediction → 0

$$\text{Precision} \rightarrow \frac{TP}{TP + FP}$$

↑↑ spam
actual → 0
prediction → 1
filtering

$$\text{f1_score} \rightarrow \frac{2 * \text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}}$$



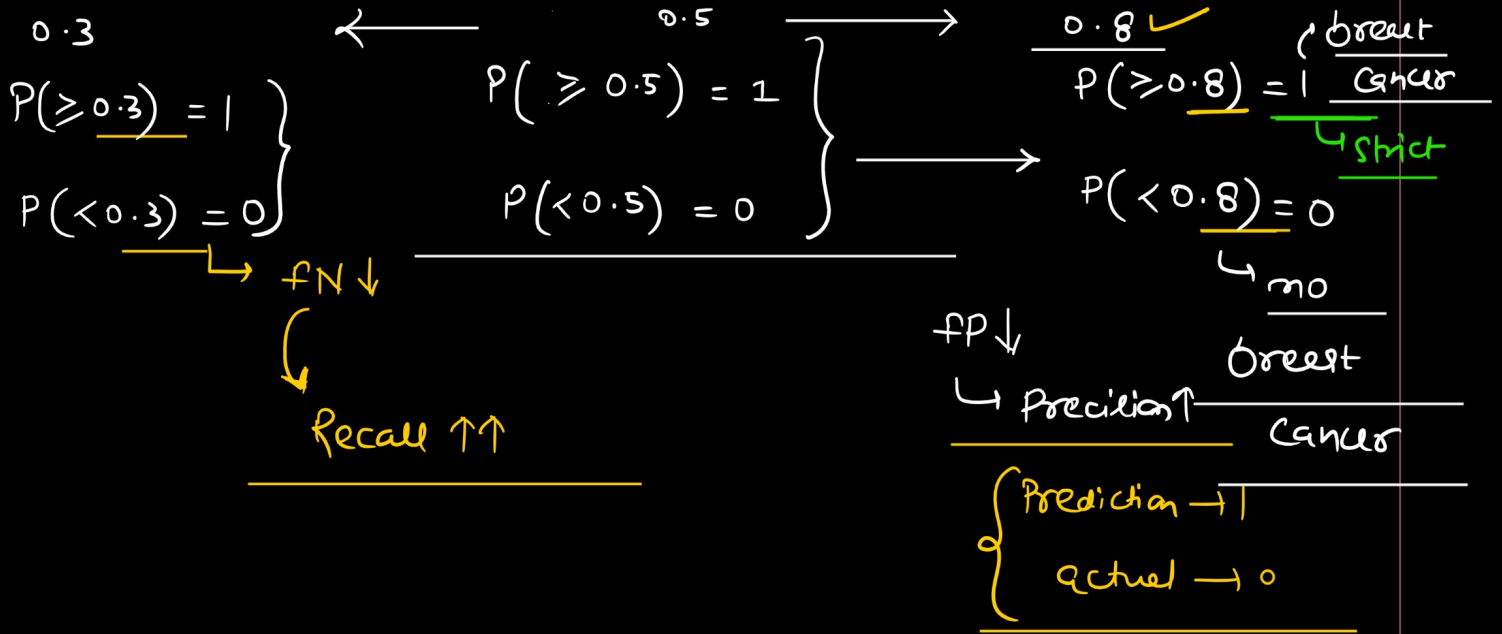
$$\frac{18}{20} \times 100 = 90\%$$

$$\frac{19}{20} \times 100 = 95\%$$

my model is
not confident to
give correct
prediction for
the diabetic
patient

$$\uparrow \text{Recall} = \frac{TP}{TP + FN} \downarrow$$

$$\uparrow \text{Precision} = \frac{TP}{TP + FP} \downarrow$$



AUC - ROC curve

$\rightarrow$ tomorrow's discussion